

Solutions and Solubility — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Every number on these papers is computed from the same solubility table the study page uses, by the generator, and checked with asserts before typesetting. The two recurring losses on this topic are the same two every year: treating "faster" and "more" as the same thing, and forgetting to scale a solubility when the water is not 100 g. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 sea water: **salt / water** [4]
tea: **sugar / water (tea)**
nail varnish: **nail varnish / propanone**
fizzy drink: **carbon dioxide / water**
One mark per row.
Row 3 is the check that "solvent" has not been learnt as "water".
- 2 (a) dissolves in a solvent [1] [3]
(b) does not dissolve [1]
(c) a solution that holds as much solute as it can at that temperature — no more will dissolve [1]
- 3 Sand: cloudy at first, then the sand settles on the bottom, water clear above [1]; a suspension [1] [4]
Salt: the salt disappears and the water stays clear / stays mixed [1]; a solution [1]
- 4 The sugar particles break away from the solid / from each other [1] [3]
and spread out between the water particles [1]
until they are evenly mixed throughout (too small to see) [1]
- 5 (a) **100 g** [1] [3]
(b) The salt particles are still in the beaker, spread between the water particles [1];
nothing has left, so mass is conserved [1]
- 6 Clear means you can see through it — the dissolved particles are too small to see [1] [2]
The blue colour shows the copper sulfate is there; clear is not the same as colourless [1]
- 7 (a) 32 g dissolves [1]; 8 g left undissolved [1] [4]
(b) Saturated [1]; undissolved solid remains on the bottom, showing the water holds as much as it can [1]
- 8 Stir it; crush it (smaller pieces / larger surface area); warm the water. One mark each. [3]
- 9 g per 100 g of water [1] [4]
Solubility changes with temperature [1]
usually rising as the water warms [1]
so the same substance has a different solubility at each temperature and the number means nothing without one [1]

Paper B — Medium

30 MARKS

- 1 Keep adding salt to the water, stirring after each addition [1] [3]
until some salt will not dissolve however long it is stirred [1]
Undissolved salt on the bottom shows it is saturated [1]
- 2 (a) 25 g is a quarter of 100 g, so $64 \div 4$ [1] = **16 g** [1] [4]
(b) 250 g is 2.5 times 100 g, so 64×2.5 [1] = **160 g** [1]
- 3 (a) **40 g** per 100 g of water [1] [7]
(b) At 20 °C: **sodium chloride** (36 g) [1]; at 60 °C: **potassium nitrate** (110 g) [1]
(c) Not saturated / unsaturated [1]; the solubility at 30 °C is 46 g, so another 16 g could dissolve [1]
(d) **Sodium chloride** [1]; its solubility only rises from 36 to 40 g across the whole range, so heating dissolves almost no more [1]
(c) uses 30 °C, a column not printed. The candidate must interpolate between 20 and 40 — 46 g is the value the page gives; accept 45–50.
- 4 110 g dissolved at 60 °C [1]; 32 g can stay dissolved at 20 °C [1] [3]
 $110 - 32 =$ **78 g** of crystals [1]
"78 g" alone is one mark. The two readings are the other two.
- 5 Solubility falls as the temperature falls, so the cooler water can hold less [1] [2]
The excess comes out of solution as solid crystals [1]
- 6 (a) Moves solvent that is already full of solute away from the solid [1] and brings fresh solvent particles into contact with it [1] [6]
(b) Increases the surface area [1], so more solute particles are in contact with water particles at once — more collisions per second [1]
(c) Water particles move faster [1], so they collide with the solid more often and with more energy, pulling particles off faster [1]
- 7 **Warming** [1] [3]
Stirring and crushing only change the rate at which the limit is reached [1]
The limit (solubility) depends only on temperature, so it is unchanged by stirring or crushing [1]
- 8 Gases (carbon dioxide) are less soluble in warm water [1] [2]
so less can stay dissolved and it escapes as bubbles more quickly [1]

Paper C — Hard

30 MARKS

- 1 (a) The solution is saturated — the water already holds the maximum it can at that temperature [1]; stirring cannot raise that limit [1] [4]
(b) It changes only the rate of dissolving [1]; it brings fresh water particles to the solid and carries dissolved ones away, so dissolving happens faster up to the limit [1]
- 2 (a) Priya: $50 - 36 =$ **14 g** [1]; Arjun: $50 - 38 =$ **12 g** [1] [5]
(b) The solubility rose by only 2 g per 100 g across 60 degrees [1]
(c) At 80 °C the water particles move faster [1], colliding with the salt more often and more energetically, so it dissolves more quickly [1]
The point of the question: heating changed the amount by almost nothing and the speed by a lot. A candidate who says heating "made no difference" in (c) has merged the two.
- 3 (a) It falls / oxygen is less soluble in warmer water [1] [3]
(b) Gas particles in the warm water have more energy [1]; so they can escape from the liquid more easily and less stays dissolved [1]

- 4** (a) $110 \div 2$ **[1] = 55 g [1]** [5]
 (b) At 20 °C, 50 g of water holds $32 \div 2 = 16$ g **[1]**; crystals = $55 - 16$ **[1] = 39 g [1]**
78 g is the answer for 100 g of water. A candidate who writes 78 has ignored the 50 g twice.
- 5** Per 100 g: $55 - 21 = 34$ g **[1]** [3]
 For 200 g: 34×2 **[1]**
= 68 g [1]
- 6** Measure a fixed mass (or volume) of water into two beakers **[1]** [5]
 Add sugar to one and salt to the other, a measured spoonful at a time, stirring until each dissolves, until no more will dissolve **[1]**
 Record the mass of each that dissolved **[1]**
 Keep the same: mass of water, temperature, stirring, time allowed **[1]**
 Sugar is more soluble if a greater mass of it dissolved in the same mass of water at the same temperature **[1]**
- 7** Saturated means the solvent holds the maximum it can; if the solubility is very small, [2]
 that maximum is small **[1]**
 so the solution holds very little solute (dilute) yet cannot hold any more (saturated) **[1]**
- 8** Nothing dissolved has disappeared — the salt particles are all still in the beaker **[1]** [3]
 12 g of **water** has evaporated into the air **[1]**
 The solution is now more concentrated; if it evaporated to dryness, all the original salt would remain **[1]**

Paper D — Hard

30 MARKS

- 1** (a) Axes labelled with units (°C, g per 100 g water) **[1]**; potassium nitrate a steep curve [8]
 rising from 13 to 246 **[1]**; sodium chloride a nearly flat line near 36–40 **[1]**
 (b) About **138 g** (accept 130–145) **[1]**; between the 110 at 60 °C and 169 at 80 °C, nearer the middle **[1]**
 (c) $169 - 64 = 105$ g **[1]** (working shown) **[1]**
 (d) $40 - 36 = 4$ g; cooling recovers almost no salt, so salt is obtained by evaporation instead **[1]**
The sodium chloride line must be drawn nearly flat. A candidate who draws both curves rising steeply has not read the data — that is the point of (a).
- 2** Award one mark for each of these, up to six: [6]
 Dissolving: sugar particles break away from the solid and spread out between the water particles.
 The water can only hold a certain mass of sugar at a given temperature — the solubility.
 When that limit is reached the solution is saturated, and any more sugar stays solid.
 Stirring only speeds up dissolving (fresh water to the solid); it cannot raise the limit.
 Warming makes the water particles move faster and collide harder, and raises the solubility.
 So at the higher temperature the water can hold more sugar, and the extra spoonful dissolves.
For full marks the answer must separate the limit (amount) from the rate (speed). Award at most four if stirring and warming are treated as doing the same thing.
- 3** (a) 150 g is 1.5 times 100 g, so 21×1.5 **[1] = 31.5 g [1]** [4]
 (b) Yes, all 30 g dissolves (30 is less than 31.5) **[1]**; total mass $150 + 30 = 180$ g **[1]**
The trap is the instinct that 30 g "sounds like more than 21 g", so some must be left. Scale the water first.

- 4** Physical: no new substance is made **[1]**; evidence: evaporate the water and the original solute comes back unchanged **[1]** [4]
Mixture: the solute and solvent are not chemically joined **[1]**; evidence: they can be separated by physical means / the proportions can vary (dilute or concentrated) **[1]**
- 5** Solubility depends on the solvent as well as the solute **[1]** [2]
so the statement must say what it is insoluble **in** — insoluble in water, but soluble in propanone **[1]**
- 6** (i) The mistake: dissolving is not melting, and mass is not lost **[1]** [6]
Correct version: the salt particles break away and spread between the water particles; every particle is still there, so the solution weighs exactly what the salt and water weighed **[2]**
(ii) The mistake: crushing changes only how fast the salt dissolves, not how much can dissolve **[1]**
Correct version: crushing gives a larger surface area, so more collisions with water particles and faster dissolving; the amount that dissolves is fixed by the temperature **[2]**

These are the two most common wrong sentences in the whole topic. If he can mark them in someone else's work, he is unlikely to write them in his own.